Mute Heart

IoT-Enabled Wearable Gesture Recognition Glove for Sinhala Sign Language Voice Translation

Final IoT Project Report | Group 19

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Introduction

The Problem

Hearing and speech-impaired individuals rely on Sri Lankan Sign Language (SSL) to communicate.

However, most of the general public does not understand SSL, creating a serious communication barrier in daily life, healthcare, and public services.

Traditional camera-based recognition systems require controlled environments and high computational power — not suitable for everyday portable use.

Our Solution

"Mute Heart" — a wearable IoT gesture recognition glove

Translates selected Sinhala sign language gestures into audible Sinhala speech in real time using:

- ESP32 Microcontroller for processing
- 5 Rotary Potentiometers for finger angles
- MPU6050 Motion Sensor for hand orientation
- DFPlayer Mini for Sinhala audio output

Currently demonstrates 5 Sinhala vowel gestures: අ, ආ, ඇ, ඉ, ඊ, උ

Sensors & Actuators

Rotary Potentiometers (×5)

Attached to each finger of the glove. Detects angular bending of fingers by producing an analog voltage proportional to the bend angle. Enables precise gesture differentiation.

MPU6050 (IMU Sensor)

6-axis inertial measurement unit capturing hand motion and orientation data (gyroscope + accelerometer). Helps identify full hand gestures beyond finger bending.

ESP32 Microcontroller

Main processing unit. Collects analog signals from potentiometers and digital data from MPU6050. Applies rule-based threshold mapping to classify gestures.

DFPlayer Mini + Speaker

Audio output module. Receives gesture commands from ESP32 and plays pre-recorded Sinhala audio files through a connected speaker to produce voice output.

Arduino Program

[Code Demonstration]

Demonstration

Thank You

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